Lab 8: Steganographic Analysis

- Course: FSCT 8561 – Security Applications
- Instructor: Dr. Maryam R. Aliabadi
- Lab Duration: 3 Hours

Overview

In this lab, you simulate an Insider Threat scenario. You are a security analyst tasked with exfiltrating footprinting data covertly. You must hide IP addresses and port scan results inside images (corporate logos or profile pictures) while ensuring minimal visual change and understanding the data capacity limitations of the images.

Learning Objectives

• Understand the binary structure of image pixels
• Use Pillow and Stepic to hide/extract data
• Analyze differences between original and stego images
• Evaluate LSB capacity and limitations

Required Reading & Tutorials

- Mastering Python for Networking and Security – Chapter 14
  - https://learning.oreilly.com/library/view/mastering-python-for/9781839217166/
  - Source code on: https://github.com/PacktPublishing/Mastering-Python-for-Networking-and-Security-Second-Edition
- Pillow (Python Imaging Library) Tutorial: https://pillow.readthedocs.io/en/stable/
- Stepic Python Module: https://pypi.org/project/stepic/

Environment Setup

1. Install required libraries:
   pip install pillow stepic
2. Download the lab assets
   a. company_logo.png
   b. profile.png
   c. python.png
   d. python_secrets.png

Part A – Manual LSB Manipulation

Task: Hide 'TARGET:192.168.1.50' inside company_logo.png using manual LSB manipulation.
Steps:

1. Run and analyze different functions of the provided code (steganography_LSB.py):
    1. `set_LSB`
    2. `get_LSB`
    3. `hide_message`
    4. `extract_message`
2. Manipulate the steganography_LSB.py file to hide 'TARGET:192.168.1.50' inside company_logo.png using manual LSB manipulation
    1. Open company_logo.png and convert to RGBA mode.
    2. Use getdata() to access pixel array.
    3. Use the provided set_LSB() function to hide the string "TARGET:192.168.1.50" in the first 10 pixels.
    4. Save the modified image as company_logo_stego.png.
    5. Save the file as IP_IN_LSB.py.
    6. Run the file.

Output:

- Python script implementing manual LSB hiding (IP_IN_LSB.py)
- Resulting stego image

Part B – Stepic

Task: Hide port scan data [80, 443, 3478, 5060] inside profile.png using Stepic.
Exfiltrate port scan results inside a profile picture using Stepic. You have performed a port scan on a conferencing tool (Zoom/Teams local instances). You need to hide the port list [80, 443, 3478, 5060] inside profile.png.

Steps:

1. Create exfiltrate.py:

```python
from PIL import Image
import stepic

# Load carrier image
carrier = Image.open("profile.png")

# Define footprinting data
footprint_data = "CONF_TOOL_SCAN: 80, 443, 3478, 5060".encode('utf-8')

# Encode and save
stego_image = stepic.encode(carrier, footprint_data)
stego_image.save("profile_secret.png")

print("Data hidden successfully.")
```

2. Run the script to generate profile_secret.png.

Part C – Verification & Analysis

Extraction: Write a corresponding decode script to recover the hidden port list (decode.py).

Visual Check: Open profile.png and profile_secret.png in a high-resolution viewer. Can you spot the difference?

File Size Check: Compare the file sizes (bytes). Does hiding data in the LSB significantly change the file size?

Deliverables

- Stego images:
  - profile_secret.png
  - company_logo_stego.png
- Python scripts:
  - LSB manual hide (IP_IN_LSB.py)
  - Stepic decode (decode.py)
- Short Analysis (PDF):
  - Screenshots of running scripts
  - Maximum number of bits for a 100x100 image using LSB
  - Why PNG is better for steganography than JPEG (lossless vs lossy)
- **Filename format:** Lab8-FirstName-LastName-StudentNumber.pdf

Grading Rubric (Total: 100 points)

Component	Full Credit	Partial Credit	Minimal/No Credit	Points
Manual LSB Script	Correctly hides message in first 10 pixels	Minor LSB errors or partial encoding	Missing or incorrect	25
Stepic Script	Correctly hides and extracts port scan data	Partial success or minor issues	Missing or incorrect	25
Verification & Analysis	Visual and file size checks complete	Partial analysis	Missing	25
Lab Report & Short Analysis and reflections	Complete, clear, correct discussion	Minor missing points	Missing or incomplete	25

Good Luck!